```
All libraries imported successfully!
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species	species_name
0	5.1	3.5	1.4	0.2	0	setosa
1	4.9	3.0	1.4	0.2	0	setosa
2	4.7	3.2	1.3	0.2	0	setosa
3	4.6	3.1	1.5	0.2	0	setosa
4	5.0	3.6	1.4	0.2	0	setosa

```
Dataset Shape: (150, 6)

Column Names:
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)', 'species', 'species_name']

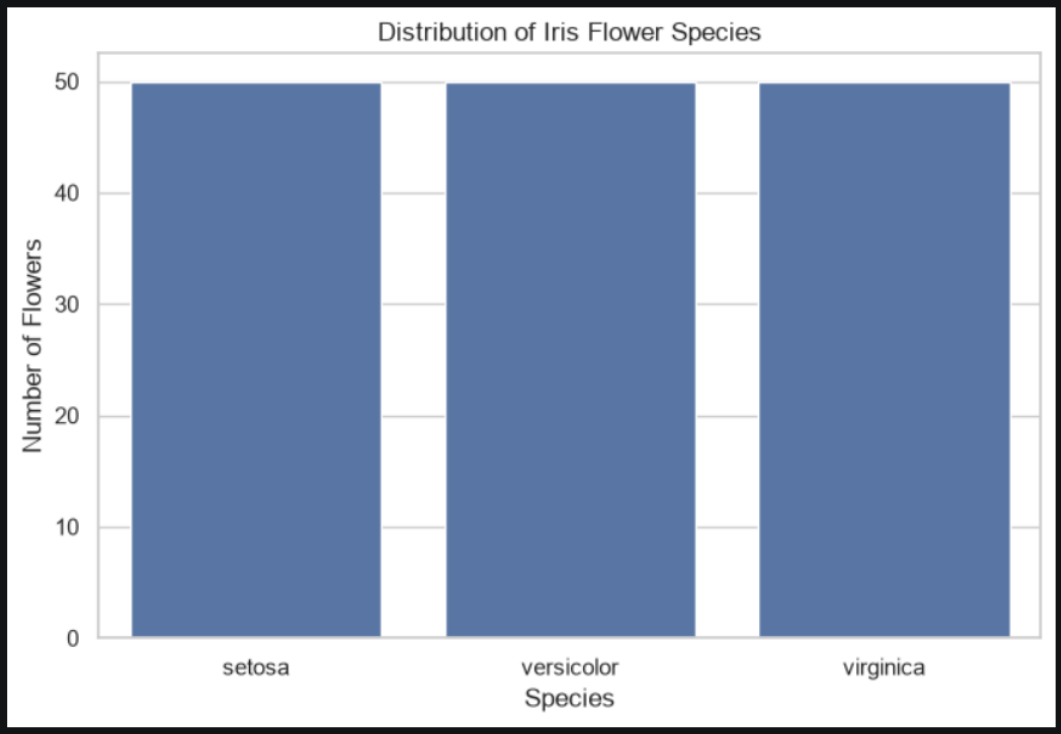
Data Types:
sepal length (cm)    float64
sepal width (cm)     float64
petal length (cm)    float64
petal width (cm)     float64
species                int64
species_name             str
dtype: object

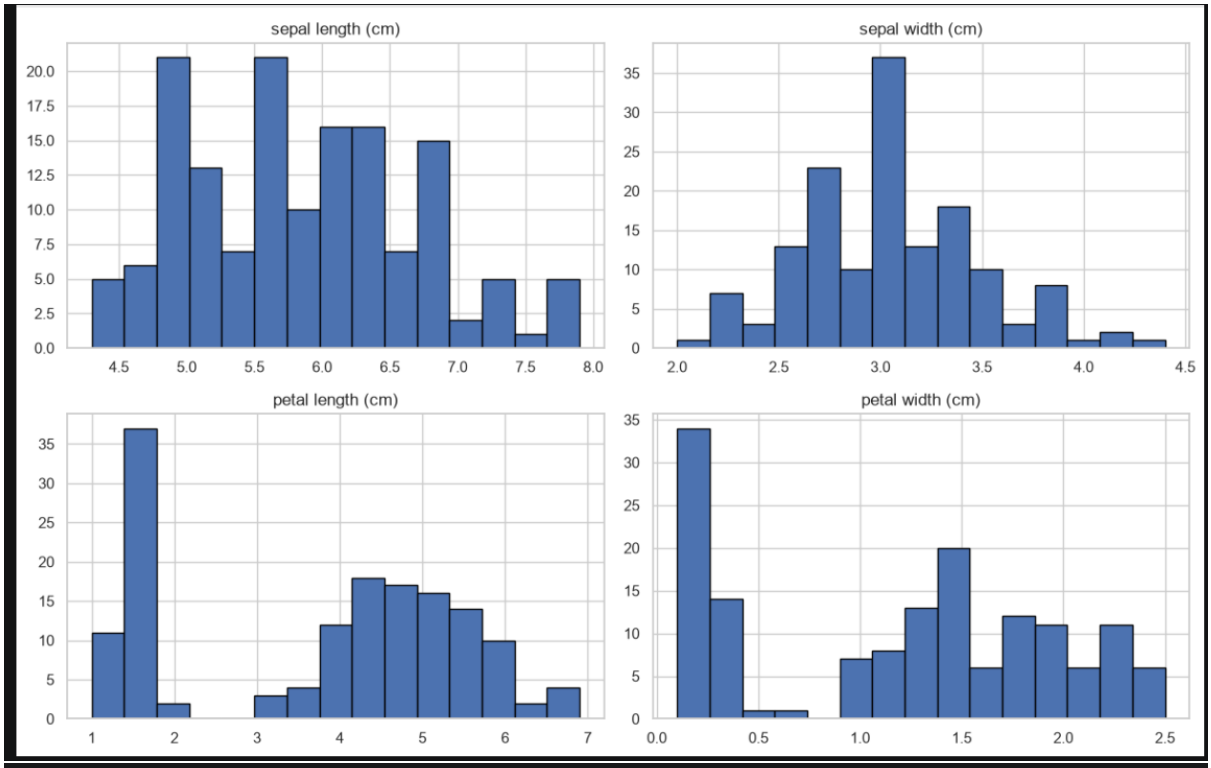
Missing Values:
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
species              0
species_name         0
dtype: int64
```

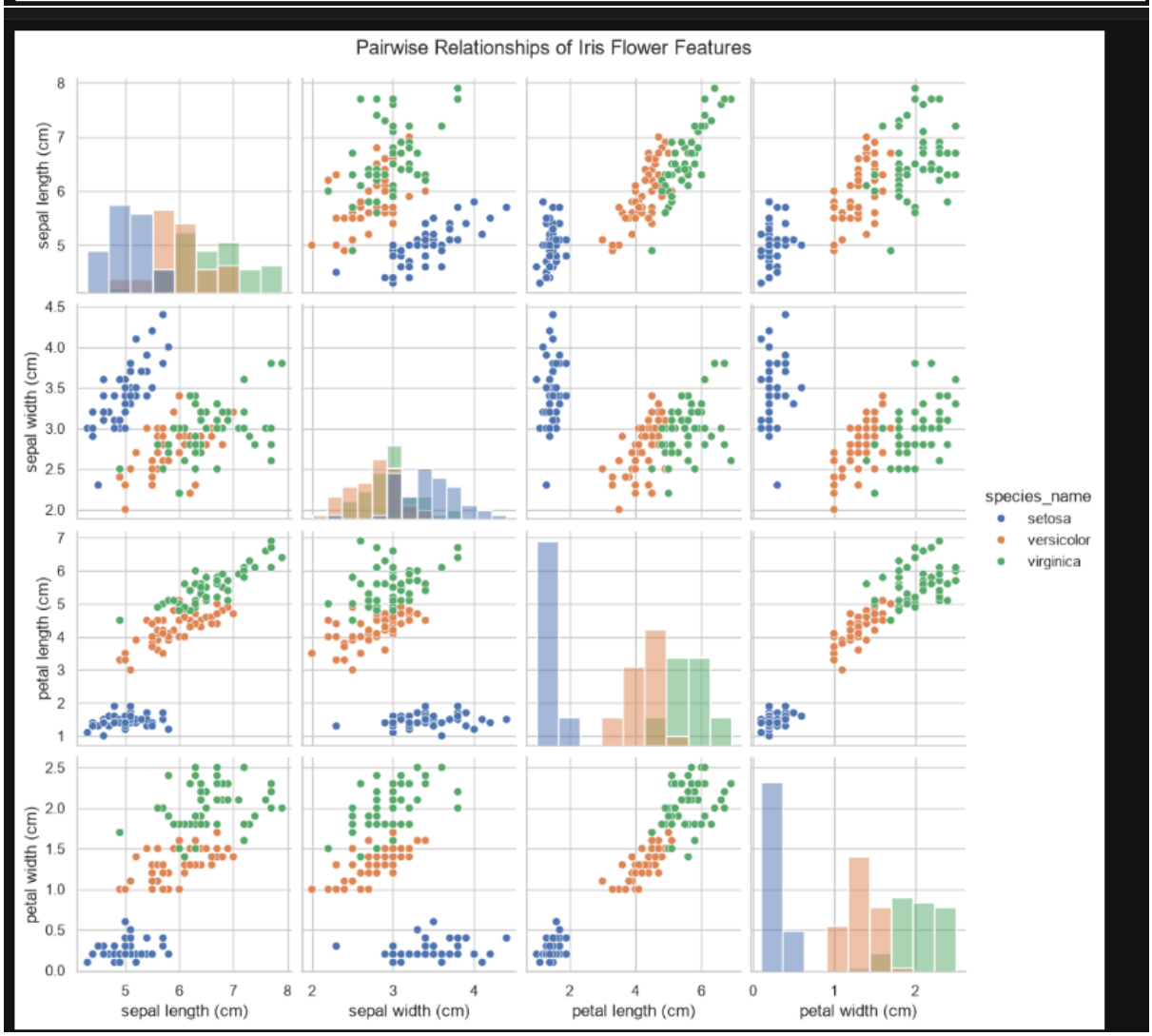
Descriptive Statistics:

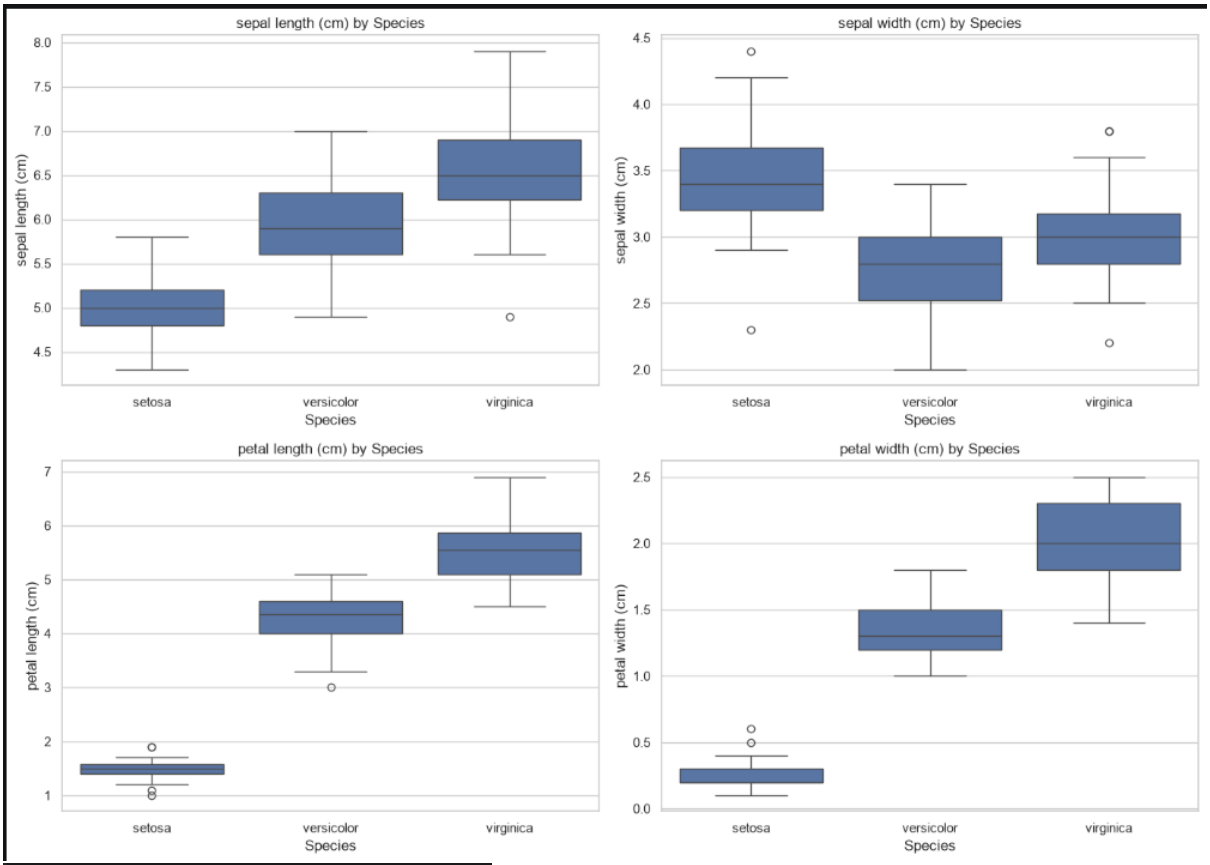
	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333	1.000000
std	0.828066	0.435866	1.765298	0.762238	0.819232
min	4.300000	2.000000	1.000000	0.100000	0.000000
25%	5.100000	2.800000	1.600000	0.300000	0.000000
50%	5.800000	3.000000	4.350000	1.300000	1.000000
75%	6.400000	3.300000	5.100000	1.800000	2.000000
max	7.900000	4.400000	6.900000	2.500000	2.000000

```
Number of flowers in each species:
species_name
setosa        50
versicolor    50
virginica     50
Name: count, dtype: int64
```




sepal length (cm)
sepal width (cm)
petal length (cm)
petal width (cm)


Pairwise Relationships of Iris Flower Features
sepal length (cm)
sepal width (cm)
petal length (cm)
petal width (cm)
species_name
setosa
versicolor
virginica




```
Training features shape: (120, 4)
Testing features shape: (30, 4)
Training target shape: (120,)
Testing target shape: (30,)
```

```
Logistic Regression model trained successfully!
```

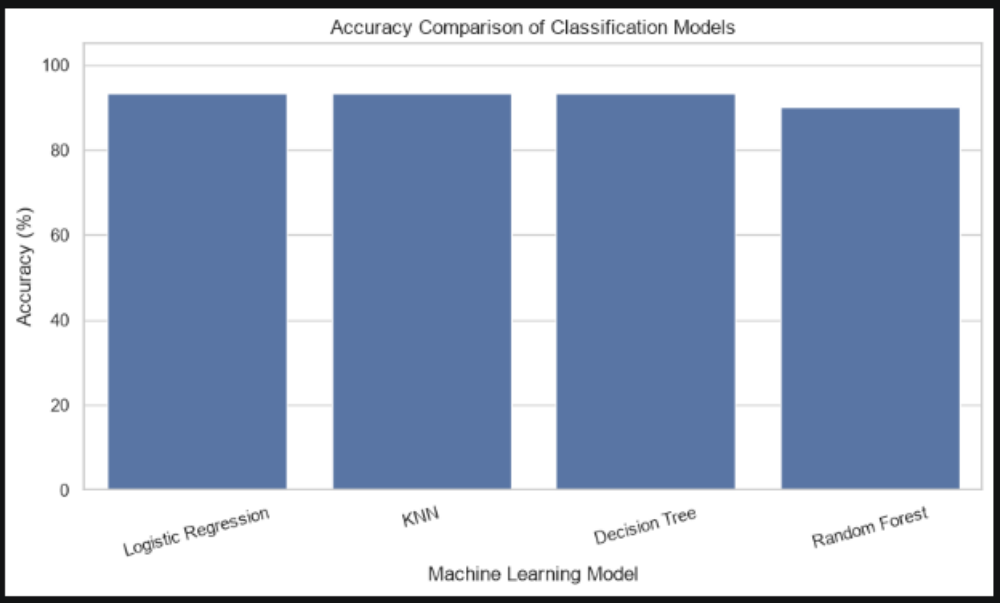
```
Decision Tree model trained successfully!
```

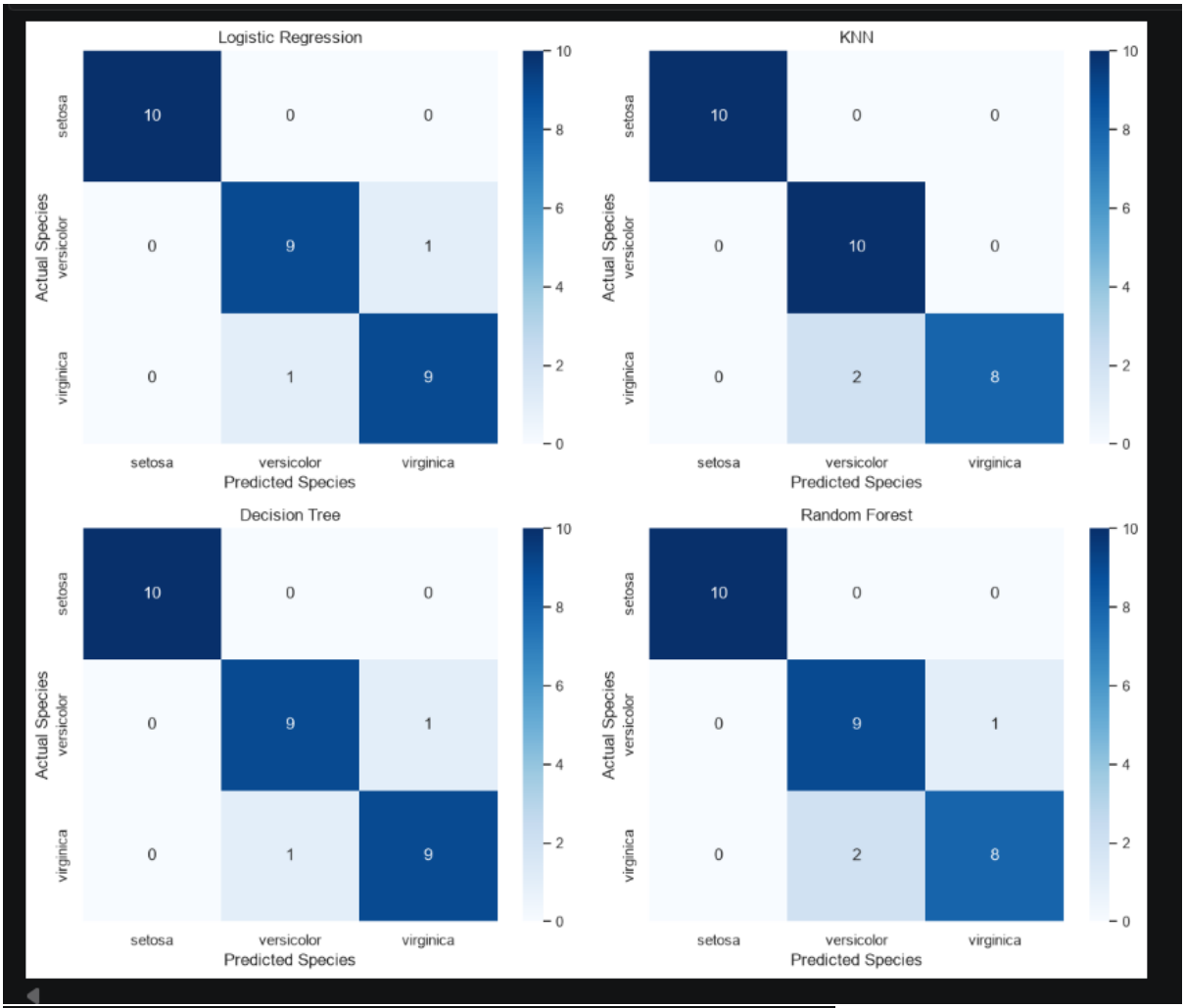
```
Random Forest model trained successfully!
```

```
Predictions generated successfully for all models!
```

Model Accuracy Comparison:

	Model	Accuracy	Accuracy (%)
0	Logistic Regression	0.933333	93.33
1	KNN	0.933333	93.33
2	Decision Tree	0.933333	93.33
3	Random Forest	0.900000	90.00






```
==================================================
Classification Report: Logistic Regression
==================================================
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       0.90      0.90      0.90        10
   virginica       0.90      0.90      0.90        10

    accuracy                           0.93        30
   macro avg       0.93      0.93      0.93        30
weighted avg       0.93      0.93      0.93        30


==================================================
Classification Report: KNN
==================================================
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       0.83      1.00      0.91        10
   virginica       1.00      0.80      0.89        10

    accuracy                           0.93        30
...
    accuracy                           0.90        30
   macro avg       0.90      0.90      0.90        30
weighted avg       0.90      0.90      0.90        30
```

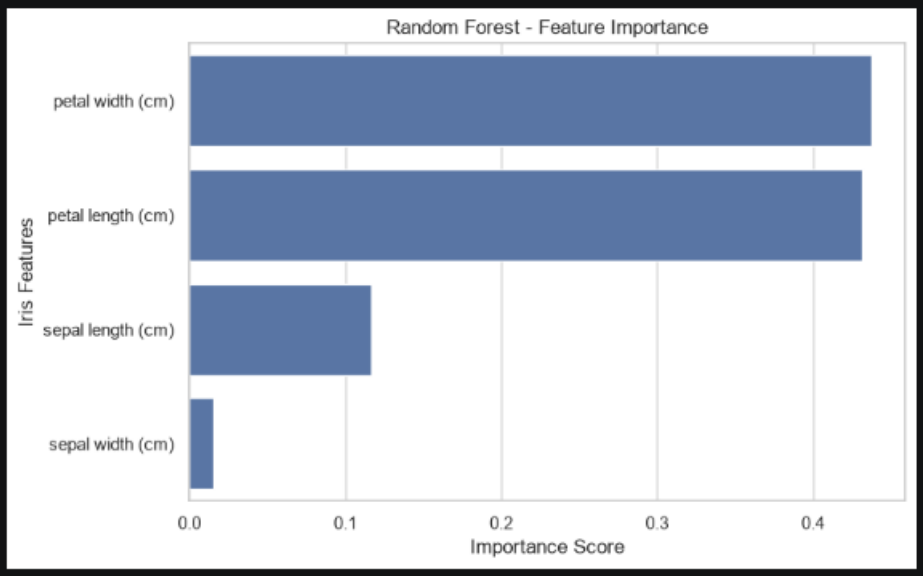
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

```
Best Performing Model: Logistic Regression
Test Accuracy: 93.33%

Reason for Selection:
The selected model achieved the highest test accuracy among the four evaluated classifiers.
```

	Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean Accuracy (%)	Std Deviation (%)
0	KNN	0.92	1.00	0.96	0.96	1.00	96.67	3.12
1	Logistic Regression	0.92	0.96	0.96	0.96	1.00	95.83	2.64
2	Random Forest	0.92	0.96	0.96	0.96	0.96	95.00	1.67
3	Decision Tree	0.92	0.96	0.96	0.96	0.92	94.17	2.04

	Feature	Importance
3	petal width (cm)	0.4372
2	petal length (cm)	0.4315
0	sepal length (cm)	0.1163
1	sepal width (cm)	0.0150




```
Total test samples: 30
Misclassified samples: 3
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	Actual Species	Predicted Species
138	6.0	3.0	4.8	1.8	virginica	versicolor
134	6.1	2.6	5.6	1.4	virginica	versicolor
77	6.7	3.0	5.0	1.7	versicolor	virginica

```
Best parameters: {'knn__n_neighbors': 5, 'knn__p': 2, 'knn__weights': 'uniform'}
Best CV accuracy: 96.67%
```

```
Test accuracy: 93.33%
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       0.83      1.00      0.91        10
   virginica       1.00      0.80      0.89        10

    accuracy                           0.93        30
   macro avg       0.94      0.93      0.93        30
weighted avg       0.94      0.93      0.93        30
```

	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
0	Logistic Regression	93.33	93.33	93.33	93.33
1	Decision Tree	93.33	93.33	93.33	93.33
2	KNN	93.33	94.44	93.33	93.27
3	Random Forest	90.00	90.24	90.00	89.97

```
FINAL MODEL COMPARISON
----------------------
Best model by weighted F1-score: Logistic Regression
Test accuracy: 93.33%
Weighted F1-score: 93.33%
```